3.7: Joining Tables of Data

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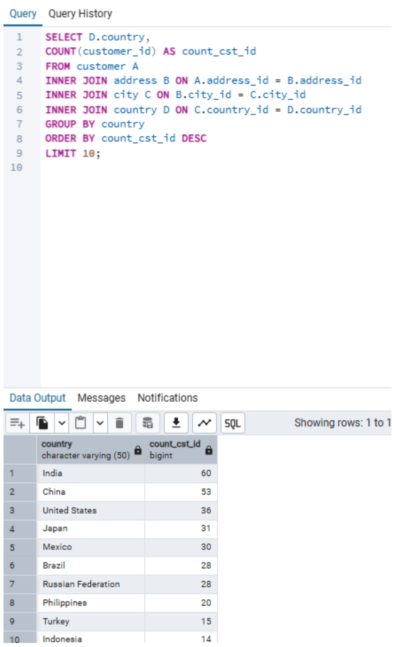
Tables: address, country, city, customer, and payment

1- 10 countries for Rockbuster in terms of customer numbers:



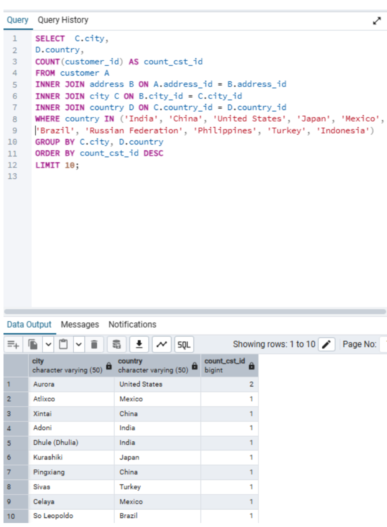

```
SELECT D.country,
COUNT(customer_id) AS count_cst_id
FROM customer A
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_id = D.country_id
GROUP BY country
ORDER BY count_cst_id DESC
LIMIT 10;
```

a. Output:

Country	Count_cst_id
India	60
China	53
United States	36
Japan	31
Mexico	30
Brazil	28
Russian Federation	28
Philippines	20
Turkey	15
Indonesia	14

b. To write this query I started by connecting common columns in the tables, and how they would connect to each other. From that, I noticed that I would need 4 tables to get from the customer count to the country column.
As every customer has a customer_id, I used this key factor as the source for my total customer per country. I also grouped by countries in order to have the total customers per each country.
Since customer_id was present on customer table, and country was on country table, I had to use multiple joins to link them (customer table -> address table -> city table -> country table).
For a more efficient query, I opted for the inner join, as I just needed limited data regarding customers count and countries. I also limited my data to 10, using a descending order, as I'm looking for the top 10 countries by customer volume.

2- Next, write a query to identify the top 10 cities that fall within the top 10 countries you identified in step 1. (Hint: the top 10 cities can be in any of the countries identified—you don't need to create a separate list for each country.)




```sql
SELECT  C.city,
D.country,
COUNT(customer_id) AS count_cst_id
FROM customer A
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_id = D.country_id
WHERE country IN ('India', 'China', 'United States', 'Japan', 'Mexico',
'Brazil', 'Russian Federation', 'Philippines', 'Turkey', 'Indonesia')
GROUP BY C.city, D.country
ORDER BY count_cst_id DESC
LIMIT 10;
```

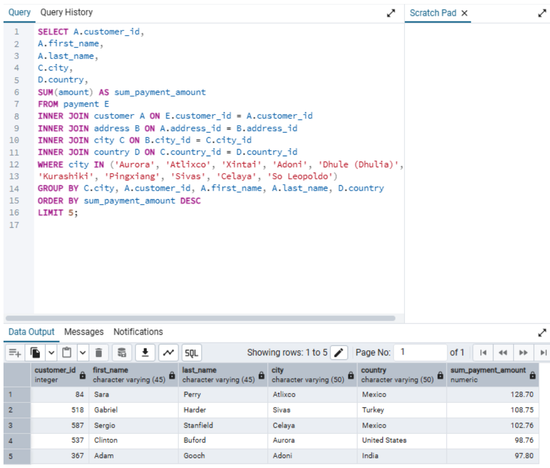
	city character varying (50)	country character varying (50)	count_cst_id bigint
1	Aurora	United States	2
2	Atlixco	Mexico	1
3	Xintai	China	1
4	Adoni	India	1
5	Dhule (Dhulia)	India	1
6	Kurashiki	Japan	1
7	Pingxiang	China	1
8	Sivas	Turkey	1
9	Celaya	Mexico	1
10	So Leopoldo	Brazil	1

```sql
SELECT  C.city,
D.country,
COUNT(customer_id) AS count_cst_id
FROM customer A
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_id = D.country_id
WHERE country IN ('India', 'China', 'United States', 'Japan',
'Mexico', 'Brazil', 'Russian Federation', 'Philippines',
'Turkey', 'Indonesia')
GROUP BY C.city, D.country
ORDER BY count_cst_id DESC
LIMIT 10;
```

To get to the 10 top cities within the 10 top countries, first I had to find out the 10 top countries, which steps were already explained on the previous question. From there, I used a new query, to retrieve limited data through multiple inner joins, so I could connect customer_id from customer table, city from city table, and country from country table.

This time, I grouped by cities and also ordered by cities in descending order, so I could get to the top cities with most customer_id count. By using the WHERE function, I was able to filter cities that were only located on the top 10 countries already outlined.

3- Top 5 customers from the top 10 cities who’ve paid the highest total amounts to Rockbuster:




```
SELECT A.customer_id,
A.first_name,
A.last_name,
C.city,
D.country,
SUM(amount) AS sum_payment_amount
FROM payment E
INNER JOIN customer A ON E.customer_id = A.customer_id
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_id = D.country_id
WHERE city IN ('Aurora', 'Atlixco', 'Xintai', 'Adoni', 'Dhule (Dhulia)',
'Kurashiki', 'Pingxiang', 'Sivas', 'Celaya', 'So Leopoldo')
GROUP BY C.city, A.customer_id, A.first_name, A.last_name,
D.country
```

Output:

customer_id	**first_name**	**last_name**	**city**	**country**	**sum_payment_amount**
84	"Sara"	"Perry"	"Atlixco"	"Mexico"	128.70
518	"Gabriel"	"Harder"	"Sivas"	"Turkey"	108.75
587	"Sergio"	"Stanfield"	"Celaya"	"Mexico"	102.76
537	"Clinton"	"Buford"	"Aurora"	"United States"	98.76
367	"Adam"	"Gooch"	"Adoni"	"India"	97.80